

When do single-cell response distributions improve drug retrieval?

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Abstract

Artificial intelligence virtual cells aim to predict how cells respond to perturbations and, ultimately, identify drugs that can drive cells towards a desired state. Transcriptomic drug-ranking methods typically summarise each perturbation by its average gene-expression response, even though single-cell measurements reveal substantial variation across cells. Whether this additional cellular variation improves drug choice remains unclear. Here we introduce population-to-population drug retrieval (PopRetrieve) to test when full single-cell response distributions improve ranking beyond average-response representations. Across multiple perturbation datasets, a magnitude-aware mean recovered most of the apparent advantage over directional mean matching, leaving a smaller and less consistent contribution from cell-to-cell response structure. This remaining benefit weakened when rankings were judged by mechanism of action or independent functional measurements. The same held-out protein measurements favoured different ranking strategies when summarised at the mean or population level, showing that evaluation resolution itself can alter the preferred method. When drug responses were directly observed in the target cellular context, full response distributions provided a small additional gain beyond average responses. When those responses instead had to be predicted in a new context, this gain was not preserved by the predictors tested here. Together, these results show that higher cellular resolution improves drug selection only when the added information changes candidate ranking and, when prediction is required, can be predicted in the target context.

Introduction

Artificial intelligence virtual cells aim to predict how cells respond to genetic or chemical perturbations and, ultimately, to identify drugs that can drive cells towards a desired state¹. Large-scale foundation models now learn reusable representations from single-cell expression atlases^{2,3}. A common strategy is to rank candidate drugs by comparing their gene-expression responses with a target response^{4,5}. Most

such approaches reduce each perturbation to an average expression signature, making large perturbation libraries tractable but discarding variation among individual cells. Single-cell perturbation atlases now retain this variation at scale⁶, revealing that the same perturbation can elicit different responses across cellular states⁷ or leave a small resistant subpopulation⁸. Whether this additional cellular detail leads to better drug choices, however, remains unclear.

Greater cellular resolution does not necessarily translate into better decisions. Two perturbations can have similar average responses yet differ substantially in how individual cells respond, so full single-cell distributions can contain information that any average representation loses. But an apparent advantage of using response populations need not arise from cell-to-cell structure itself. Conventional directional matching discards response magnitude, whereas many population-level comparisons retain it, allowing gains that can be explained by the average response alone. Moreover, the value of a richer representation depends on how success is defined. Recent benchmarks assess perturbation-response prediction across cellular contexts and generalisation scenarios⁹. Better agreement with a transcriptomic response distribution does not necessarily imply better recovery of biological mechanism or functional similarity. The relevant question is therefore not simply whether single-cell measurements contain more information, but whether the additional information changes the ranking of candidate drugs under an evaluation that reflects the intended use.

This question becomes more important in prospective applications, where the responses of candidate drugs in a new cellular context are not observed and must first be predicted^{10,11}. Existing models predict unseen drug combinations and doses¹², responses to previously unseen compounds¹³, and outcomes of multigene perturbations¹⁴. Cell-type-specific drug-response prediction has also been explored in limited-data settings¹⁵. Population-level ranking can benefit from cell-to-cell response structure only if the predictor preserves the perturbation-dependent variation that matters for drug choice. A model may reproduce heterogeneous cells without accurately capturing how different cellular states respond to the perturbation. Thus, the value of higher cellular resolution depends not only on whether additional response structure exists, but on whether it changes candidate ranking and can be recovered in a new context.

Here we introduce population-to-population drug retrieval (PopRetrieve) to determine when single-cell response distributions improve drug ranking beyond average-response representations. We separate information carried by response direction, response magnitude and cell-to-cell structure, and test whether the resulting gains persist under biological and functional evaluation and when candidate responses must be predicted. Across multiple perturbation datasets, a magnitude-aware mean recovers most of the apparent advantage of population-level retrieval, whereas the additional contribution of cell-to-cell structure is smaller and depends on how success is evaluated. Population structure can alter drug choice when candidate responses are observed, but this advantage is not preserved by the predictors tested here. Together, these results define when higher cellular resolution provides information that is not only measurable, but consequential for intervention choice.

Results

Population retrieval retains response information beyond mean signatures

Population-to-population retrieval ranks candidates by comparing query and candidate response populations directly (Fig. 1a,b), whereas classical mean-signature retrieval^{4,5} first collapses each population to a single mean differential-expression vector. Retaining the cells makes response structure available that averaging removes, including differences between cellular states or subpopulations. The ranking

task is the same; the representation determines how much measured response reaches the decision. Two candidate populations can share the same mean response and still differ markedly in how individual cells respond (Fig. 1c,d). A population-aware score can separate such candidates because it retains distributional structure that any mean-only score discards. Whether that additional information matters for retrieval is a decision question: it matters only when it changes relative candidate scores enough to alter candidate ordering (Fig. 1e). This distinction between information retained and information that changes the ranking motivates the analyses that follow. These representations form an information ladder: directional matching retains response orientation, magnitude-aware means additionally retain response strength, and population comparison can also retain cell-to-cell response structure (Fig. 1a,b). For energy distance, population comparison reduces to mean- L_2 ranking as within-population variation vanishes under a common reference (Methods). Subpopulation-coverage scores additionally test whether candidates match all query states (Supplementary Table 1).

A magnitude-aware mean recovers most gains over directional matching

Population-level retrieval substantially outperformed directional matching, but most of the gain did not require the full response population. Across seven controlled-mixture task-setting combinations with observed candidate responses, energy-distance retrieval reached a macro-averaged Hit@1 (the fraction of queries with the correct candidate ranked first) of 0.836, compared with 0.388 for mean cosine and 0.788 for magnitude-aware mean L_2 (Fig. 2a). The mean- L_2 comparison changes both magnitude sensitivity and score geometry; it is therefore a magnitude-aware baseline rather than a pure magnitude ablation. The additional distributional benefit varied across tasks. In Frangieh Perturb-CITE-seq¹⁶, magnitude-aware mean L_2 performed best, and energy distance fell below both mean baselines (Fig. 2b). Under partial observation, population scores again improved on directional matching, but not on the magnitude-aware mean. Worst-case subpopulation coverage reduced response-matching regret by a median of +0.0304 relative to mean cosine (95% drug-cluster bootstrap CI +0.0155 to +0.0495; Fig. 2c). Regret measures the loss in response matching relative to the best candidate in the available library (Methods).

The energy-distance advantage over directional matching also varied with minority-state abundance and cell line (Fig. 2d). Across five population scores, all improved on mean cosine, whereas none showed a positive median gain over magnitude-aware mean L_2 under partial observation (Fig. 2e). The qualitative conclusions were unchanged after matched-control referencing, although individual rankings could change under partial observation (Supplementary Note 1). Mean- L_2 scores also correlated more strongly with population scores than with directional scores (Fig. 2f). A magnitude-aware mean score therefore recovers most of the gain beyond directional scoring, while the additional benefit of the full population is smaller and not consistently supported across evaluation settings. Because both criteria reward response correspondence, the next analysis tested whether the increment survives biological and functional evaluation.

Population-level gains depend on how success is evaluated

The value of that information depends on the evaluation criterion. We kept rankings fixed and changed only the criterion used to judge them (Fig. 3). Response-matching gains did not consistently translate into recovery of drugs with the same mechanism of action. Across 600 leave-drug-out queries, worst-case subpopulation coverage improved response matching over mean cosine by a median of +0.028 (95% cluster-bootstrap CI +0.0114 to +0.0510). The same rankings gave a median mechanism-recovery gain

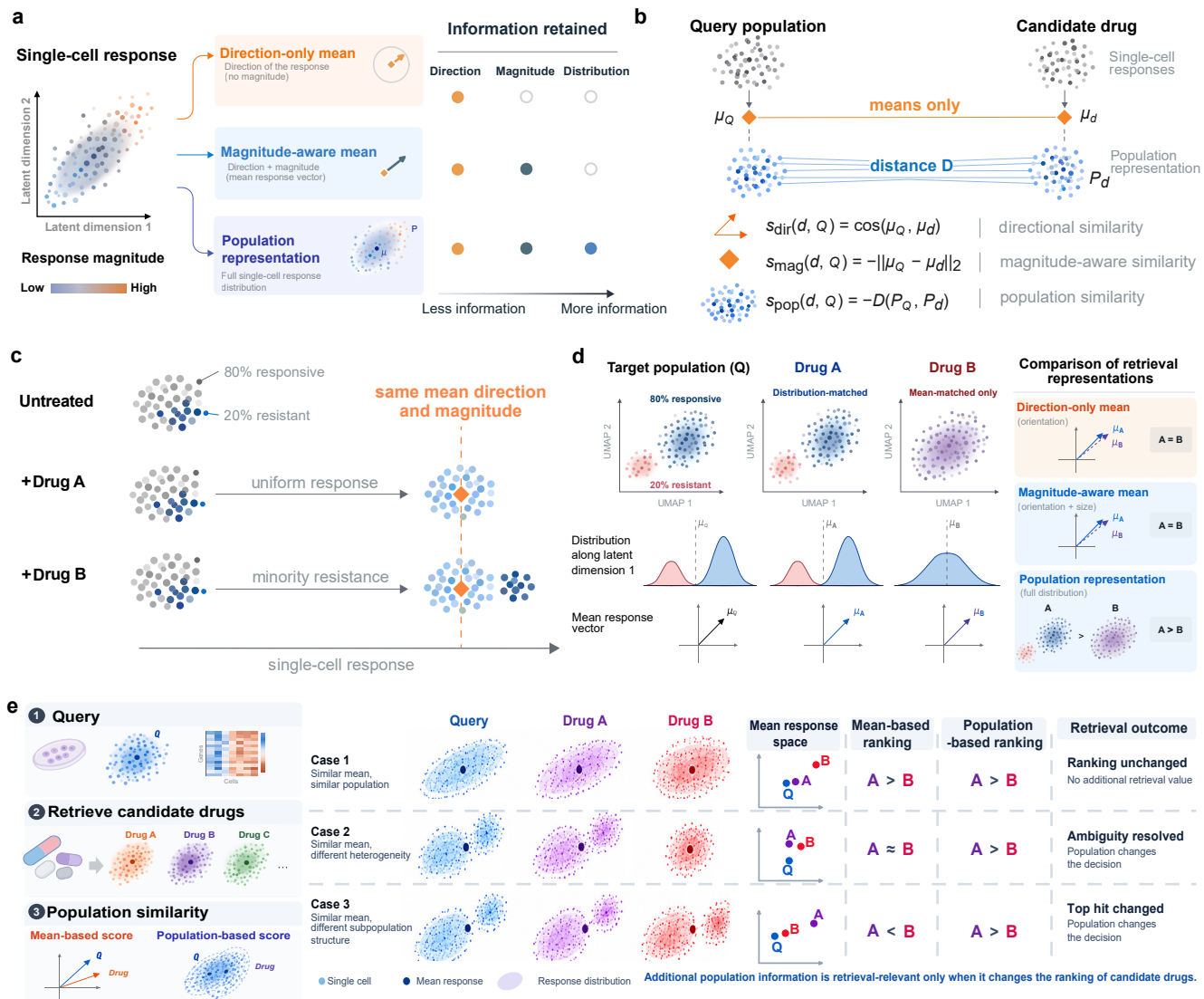


Figure 1: Single-cell response representations retain different information and can change candidate ordering. **a**, Three representations of the same single-cell perturbation response retain progressively more information: response direction only, direction and magnitude of the control-referenced mean response, and the full raw treated-cell response population. Filled and open circles indicate retained and discarded information. **b**, Corresponding retrieval rules compare query and candidate responses using directional similarity, magnitude-aware distance between control-referenced mean responses, or a distance between raw treated-cell response populations. **c**, Two perturbations can have the same mean response direction and magnitude while differing in their single-cell response structure. Drug A produces a relatively uniform response, whereas drug B leaves a resistant minority population. **d**, A target population and two candidates illustrate the resulting ambiguity. Drug A matches the target distribution, whereas drug B matches only its mean; the direction-only and magnitude-aware mean scores are therefore tied, while the population representation prefers drug A. **e**, Population information affects retrieval only when the additional response structure changes candidate ordering. Across three illustrative cases, population-based scoring either preserves the mean-based top-ranked candidate or resolves ambiguities that are invisible to mean-based matching.

of 0.000 and a mean of -0.024 (Fig. 3a,b). All five population scores showed this divergence between response matching and mechanism recovery (Supplementary Note 1). A mean-based minority-state proxy failed to track cell-level preservation and was excluded from biological interpretation (Fig. 3c). Response divergence did not reliably identify positive mechanism-recovery gains, and a pre-specified selection rule did not enrich for response-matching benefit (Fig. 3d-f; Supplementary Note 1).

Population-level retrieval also showed an association with the external functional readout. Using GDSC2^{17,18} dose-response profiles from cell lines excluded from the transcriptional retrieval experiment, and removing the shared cell-line sensitivity effect, energy-distance retrieval reached a median per-query Spearman correlation of $+0.265$ with drug-drug functional similarity, against $+0.083$ for

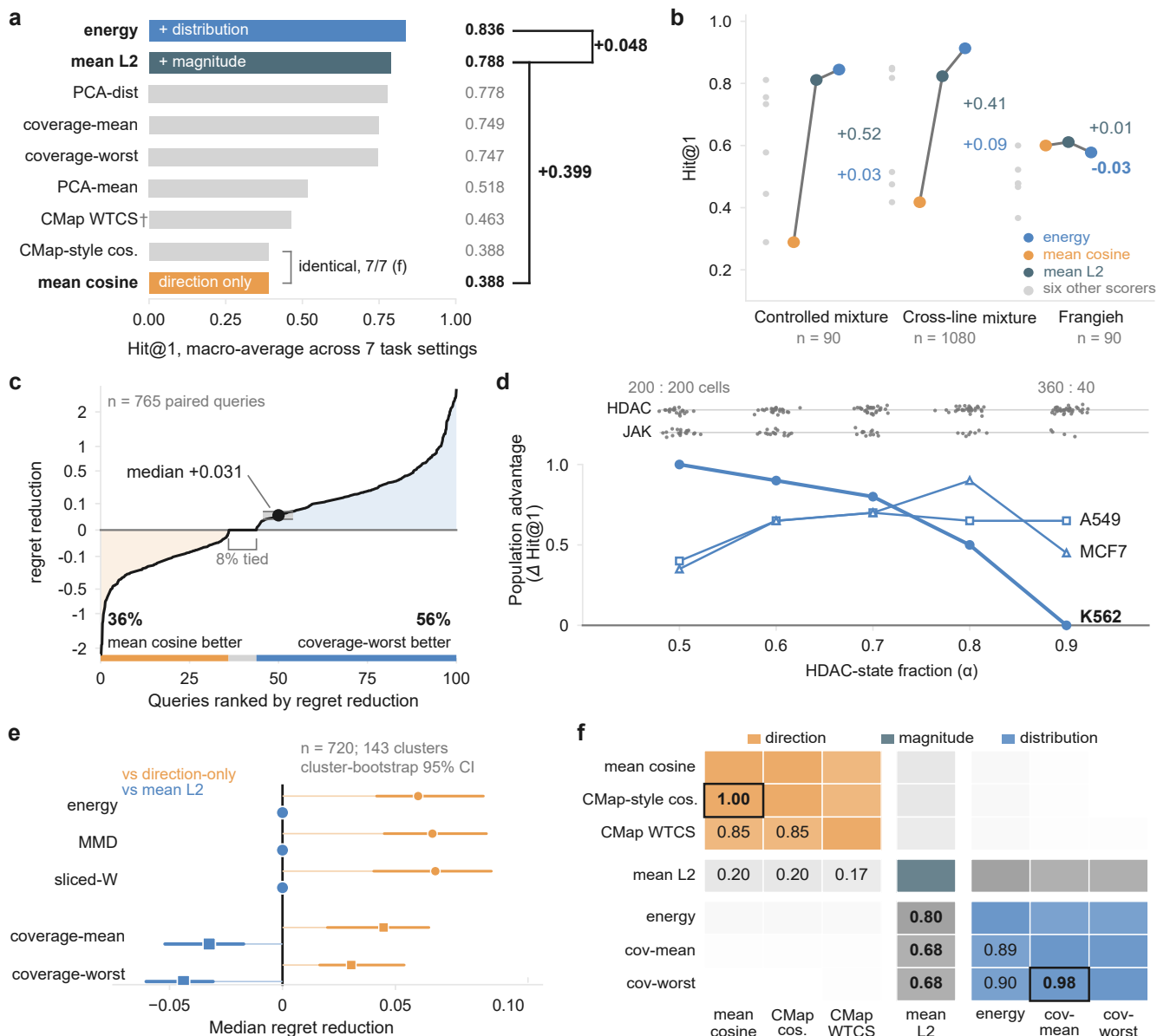


Figure 2: A magnitude-aware mean score recovers most of the gain beyond directional scoring. **a**, Hit@1 for nine scores, macro-averaged over seven task-setting combinations. Direction-only mean cosine, magnitude-aware mean L2 and energy distance reach 0.388, 0.788 and 0.836, respectively. Dagger, published CMap connectivity score (WTCS). **b**, The three scores by task ($n = 90, 1,080$ and 90 queries). The magnitude step is $+0.52$ and $+0.41$ for the two SciPlex3 tasks and $+0.01$ for Frangieh; the distributional step for Frangieh is -0.03 . **c**, Worst-case subpopulation coverage versus direction-only mean cosine for 765 partial-observation queries. The curve shows the quantile function of per-query regret reduction, with positive values favouring coverage. Formal inference used 720 queries from 143 query-drug clusters and gave a median reduction of $+0.0304$ (95% cluster-bootstrap CI $+0.0155$ to $+0.0495$). **d**, Energy advantage over mean cosine as a function of minority fraction α ($n = 20$ seeds per point). In K562 the advantage falls from $+1.00$ to 0.00 ; in A549 and MCF7 it ranges from $+0.40$ to $+0.65$ and from $+0.35$ to $+0.45$, respectively. **e**, Median regret reduction for five population-level scores across all 720 individual-drug queries from 143 query-drug clusters. All five scores improve on direction-only mean cosine (all 95% cluster-bootstrap CIs exclude zero). Relative to magnitude-aware mean L2, the three distributional scores have zero median regret reduction with CIs including zero, whereas the two coverage scores are negative (-0.030 and -0.041). **f**, Spearman correlations among seven scores for 54,180 query-candidate comparisons from 1,260 queries. Shade, $|\rho|$; rings, the closest pair within a tier. Mean L2 correlates more strongly with distributional scores ($\rho = 0.68\text{--}0.80$) than with direction-only scores ($\rho = 0.17\text{--}0.20$).

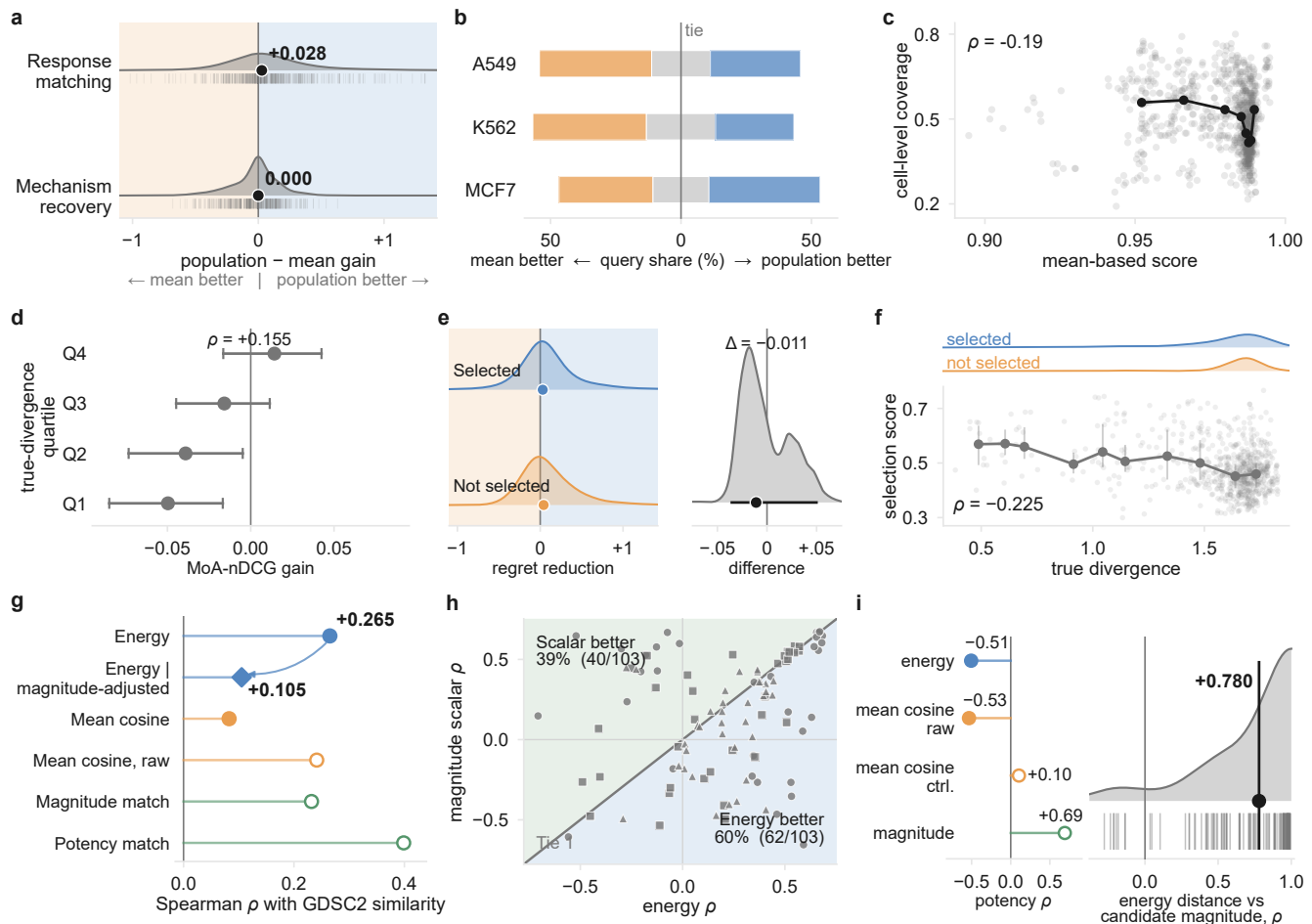


Figure 3: The population advantage weakens, and can change direction, under biological and functional evaluation. **a**, 600 leave-one-drug-out queries from 130 query-drug clusters, evaluated by response matching and mechanism recovery. Black points and thick lines show medians and 95% cluster-bootstrap CIs: +0.028 (+0.0114 to +0.0510) and 0.000 (0.0000 to 0.0000), respectively. **b**, Directional composition of mechanism-recovery gain for 200 paired queries per cell line. Warm, grey and cool segments denote decreased, tied and increased recovery under population retrieval, respectively: 86/45/69 in A549, 87/53/60 in K562 and 72/43/85 in MCF7. Across one median effect per held-out drug ($n = 130$), two-sided Wilcoxon $P = 0.119$. **c**, Comparison of a mean-based minority-state similarity score with a cell-level nearest-neighbour measure in 860 candidate cases from 20 controlled-mixture seeds. Grey points are candidate cases; the black line joins medians in eight equal-count bins. Pooled Spearman $\rho = -0.188$; the two measures selected different top-ranked candidates in 18 of 20 seeds. **d**, Population-minus-mean MoA-nDCG gain by response-divergence quartile. Points and intervals are quartile means and 95% query-drug-cluster bootstrap CIs; cluster-bootstrap Spearman $\rho = +0.1548$ (95% CI +0.0722 to +0.2349). **e**, Regret-reduction distributions for 584 queries selected by a pre-specified rule and 125 unselected queries; right, their cluster-bootstrap median difference across 141 query-drug clusters (-0.0109 , 95% CI -0.0355 to $+0.0498$). **f**, Selection score versus measured response divergence among 720 individual-drug queries ($\rho = -0.2247$, 95% cluster-bootstrap CI -0.3144 to -0.1289); top, divergence distributions for selected and unselected queries. **g**, Median ranking correlations with GDSC2 drug-drug functional similarity ($n = 103$). Stems begin at zero; the arc links energy retrieval to its magnitude-adjusted value. **h**, Per-query comparison of energy distance with the magnitude-matching scalar; energy wins on 62 queries, the scalar wins on 40, and one query ties. **i**, Left, median ranking correlations with absolute GDSC2 potency: energy and raw mean cosine are negative (-0.506 and -0.533), whereas control-subtracted mean cosine and magnitude are positive ($+0.105$ and $+0.692$). Right, per-query Spearman correlation between energy distance and candidate magnitude; median $+0.780$.

the corresponding mean-signature baseline (Fig. 3g; Supplementary Table 3). Response magnitude accounted for much of this association. Ranking candidates only by how closely their overall response magnitude matched the query reached +0.232 from a single scalar per perturbation, and once that magnitude channel was accounted for the remaining association for energy retrieval was +0.105, positive in all three cell lines (Fig. 3g,h). Scored against absolute GDSC2 potency instead, energy and raw mean cosine turn negative, since matching a weak query's magnitude returns other weak drugs, whereas ranking by candidate magnitude alone turns positive (Fig. 3i). The channel is also directly measurable: across queries the query-candidate energy distance tracks the candidate's own response magnitude at a me-

dian Spearman correlation of +0.780 (Fig. 3i). Population-level retrieval was therefore associated with external functional similarity, but much of this association was recoverable from response magnitude. The value of population-level retrieval therefore depends on how success is measured, including criteria beyond transcriptomic response matching.

Evaluator resolution changes the preferred retrieval strategy

We tested this dependence using a biological measurement unavailable to the retrieval methods. Frangieh Perturb-CITE-seq¹⁶ measures RNA and 20 surface proteins in the same 218,331 cells, allowing RNA-derived rankings to be judged using protein responses withheld from every retrieval method. To separate evaluator resolution from score geometry, we constructed parallel mean- and population-level evaluators from the same protein measurements. The mean evaluator used the negative Euclidean distance between control-referenced mean protein responses, whereas the population evaluator used the negative energy distance between the corresponding protein-response populations. We then evaluated two fixed RNA-derived rankings from the same representation ladder: magnitude-aware mean-L2 retrieval and population-level energy retrieval (Fig. 4a).

Under the protein mean-L2 evaluator, magnitude-aware mean RNA retrieval performed better than population RNA retrieval, with median per-query Spearman correlations of +0.4255 and +0.4150, respectively; the median paired difference, energy minus mean-L2, was -0.0208 . Under the protein population-energy evaluator, the ordering reversed: population RNA retrieval reached +0.3930 against +0.3199 for magnitude-aware mean retrieval, with a median paired difference of +0.0538 (Fig. 4b). The paired difference had the same sign in all three immune conditions: -0.0105 versus +0.0466 in Co-culture, -0.0363 versus +0.1342 in Control, and -0.0318 versus +0.0384 in IFN γ for the mean and population evaluators, respectively (Supplementary Table 5). Thus, the same held-out protein measurements favour different RNA-side retrieval strategies depending on whether the evaluator retains only the mean response or the full response population, even when score geometry is matched.

Response magnitude alone explained part, but not all, of this dependence. A scalar ranking that compares only the overall magnitude of the query and candidate RNA responses reached median correlations of +0.3223 under the protein mean-L2 evaluator and +0.2764 under the protein population-energy evaluator, compared with +0.1206 under the direction-only protein-cosine reference. These values remained below the corresponding matched retrieval strategy under each primary evaluator: +0.4255 for magnitude-aware mean RNA retrieval under the mean evaluator and +0.3930 for population RNA retrieval under the population evaluator (Fig. 4c). The evaluator-form reversal therefore cannot be reduced to response magnitude alone.

State-specific responses can be recoverable without reordering candidates

HIR-Bench provides a controlled example in which different population-level objectives can favour different candidates, with an analytical comparison between majority- and minority-specialist candidates (Methods; Fig. 4d). Its exploratory transfer analysis used a mean-based minority-state proxy that did not support biological interpretation, so it does not establish a boundary for biological retrieval (Supplementary Note 2). State-dependent perturbation responses were recoverable in naturally heterogeneous glioblastoma tissue⁷. Malignant and myeloid compartments were analysed separately against compartment-matched controls. A supervised probe recovered within-patient drug-response identity more accurately than the best unsupervised procedure, with a median paired gap of +0.117 (Fig. 4e; Supplementary Note 3).

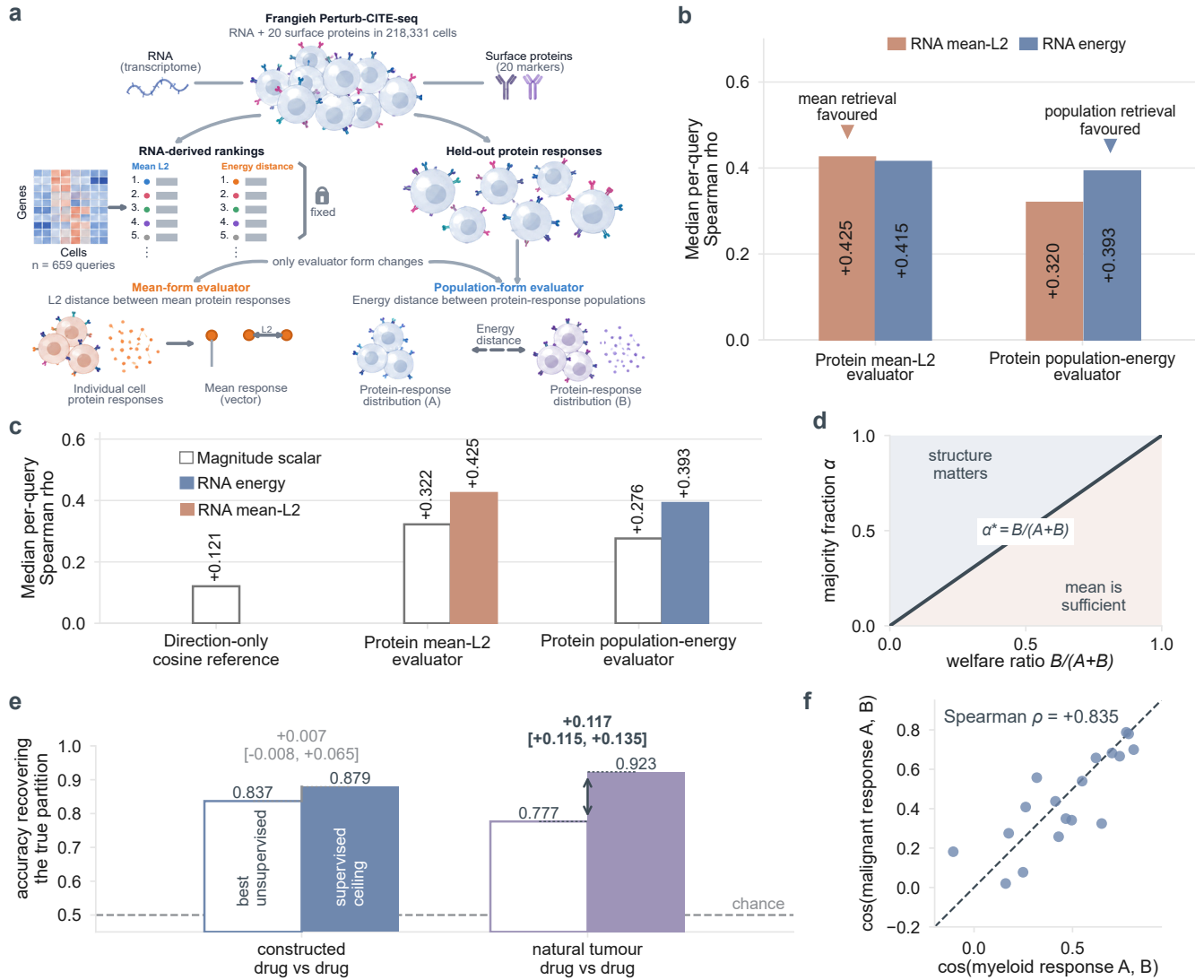


Figure 4: External evaluator form reverses the preferred retrieval strategy under matched geometry, whereas heterogeneity need not reorder candidates. a–c, The same fixed RNA-derived rankings evaluated using held-out protein responses from Frangieh Perturb-CITE-seq. **a**, RNA and 20 surface proteins are measured in the same 218,331 cells; protein measurements are unavailable to the retrieval methods. Two primary protein evaluators are constructed from the same cells: a magnitude-aware mean evaluator using negative Euclidean distance between control-referenced mean protein responses, and a population evaluator using negative energy distance between protein-response populations. **b**, Median per-query Spearman correlation across $n = 659$ queries for magnitude-aware mean-L2 and population-energy RNA retrieval. Under the protein mean-L2 evaluator, RNA mean-L2 and RNA energy reach +0.4255 and +0.4150, respectively, with a median paired difference of -0.0208 (energy minus mean-L2). Under the protein population-energy evaluator, the ordering reverses: +0.3199 and +0.3930, with a median paired difference of +0.0538. The paired difference has the same sign in Co-culture (-0.0105 , +0.0466), Control (-0.0363 , +0.1342) and IFN γ (-0.0318 , +0.0384) for the mean and population evaluators, respectively. **c**, A scalar RNA response-magnitude ranking reaches +0.3223 and +0.2764 under the protein mean-L2 and population-energy evaluators, respectively; its value under the direction-only protein-cosine reference is +0.1206. **d**, HIR-Bench boundary $\alpha^* = B/(A+B)$ separating settings in which the mean is sufficient from those in which subpopulation structure can change the preferred candidate. **e**, **f**, Four glioblastoma tumours analysed in disjoint malignant and myeloid compartments with matched drugs and controls. **e**, Recovery of within-patient drug-response identity; open bars, best unsupervised method; filled bars, supervised probe. Median paired gap, +0.117 (95% patient-level cluster-bootstrap CI +0.115 to +0.135, $n = 36$ splits), versus +0.007 in the constructed comparator. **f**, Myeloid-response similarity versus malignant-response similarity across compartments (Spearman $\rho = +0.835$, $n = 18$ drug pairs); dashed line, equal similarity.

Whether that structure is large enough to change which drug should be preferred is a separate question, and the tumour data constrain it. Across 18 within-patient drug pairs, myeloid-response similarity ranked malignant-response similarity at Spearman $\rho = +0.835$ across the two disjoint compartments, which share no cells (Fig. 4f). The coefficient describes this cohort. Drug-response similarity was largely preserved across distinct cellular compartments, leaving less room for state-specific structure to reorder candidates than the presence of heterogeneity alone would suggest. This measures preservation of drug ordering; the intervention-identification task below tests whether cellular structure changes candidate preference. The same pattern held at larger scale in Tahoe-100M. State-specific perturbation responses remained recoverable (supervised ceiling 0.853; best unsupervised procedure 0.611), yet relative drug ordering was largely preserved across independently defined cellular states (median disjoint Spearman $\rho = 0.841$ for cell-cycle states and 0.778 for control-derived states; Supplementary Note 3).

Observed candidate responses reveal a small population-level gain beyond the mean

Prospective drug retrieval ultimately requires ranking compounds whose effects in the target context have not been measured. We first established the information ceiling by making those candidate responses directly available, before replacing them with predictions in Fig. 6. In the Tahoe-100M intervention-identification task¹⁹, untreated cells and a desired treated population define a query for ranking a fixed 92-compound library (Fig. 5a). We evaluated the true intervention’s rank across 3,992 queries from 44 cell lines. The observed and predicted arms separate the value of population information from the cost of forecasting it. With candidate responses observed, mean reciprocal rank (MRR), which increases as the true intervention moves towards the top of the ranking, increased from 0.945 for directional mean cosine to 0.965 for magnitude-aware mean L_2 and 0.976 for energy-population retrieval (Fig. 5b). The additional population-level gain depended on the score: maximum mean discrepancy reproduced it, whereas sliced Wasserstein distance performed below both mean baselines (Supplementary Note 1). Energy-population retrieval also changed individual decisions. Relative to magnitude-aware mean L_2 , it corrected the top-ranked drug on 2.0% of queries and broke a correct choice on 0.5% (Fig. 5c).

The oracle gain was concentrated in queries for which directional mean matching left room to improve (Fig. 5d). A ceiling-preserving permutation null nevertheless showed that headroom alone did not identify population-beneficial cases (Fig. 5e; Methods). Among measured query properties, response detectability was associated with population-route performance after conditioning on directional mean-cosine retrieval. Neither cross-fitted drug-by-state interaction nor its reproducibility showed a supported association (Fig. 5f–h; Methods). Population information therefore had measurable value with observed responses, but limited room to improve retrieval.

The observed-response population advantage is not preserved by the evaluated predictors

Replacing the observed candidate responses with predicted ones removes the advantage and reverses its sign (Fig. 6a). The primary average-effect predictor costs direction-only mean cosine 0.058 MRR and energy-population retrieval 0.124, and the two cross (Fig. 6b). Under it energy-population retrieval sits -0.0356 below direction-only mean cosine (95% CI -0.0620 to -0.0115), so the oracle gain does not attenuate but changes sign (Fig. 6c). None of the six forward predictors shows a statistically supported population advantage; the Gaussian optimal-transport map is the only one with a positive point estimate ($+0.0027$, interval covering zero). The dominant deficit is a magnitude error rather than a distribution-specific one. On the same predictions the magnitude-aware mean loses -0.0335 to direction-only mean

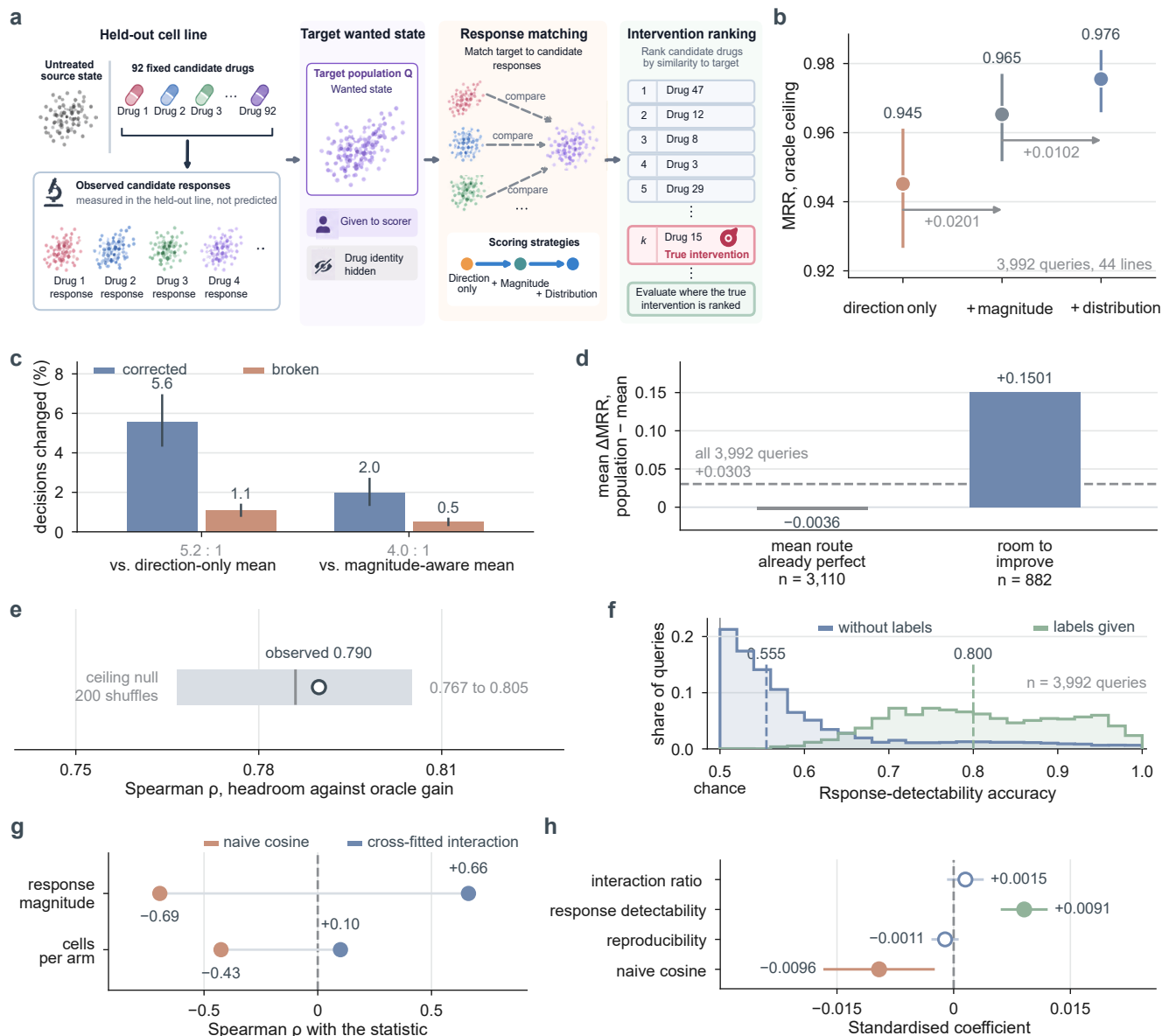


Figure 5: Population structure adds a small oracle retrieval gain beyond response magnitude. Source-to-target intervention identification on Tahoe-100M: 3,992 queries over 44 cell lines and a fixed 92-compound library, with candidate responses observed within the query line; predicted candidates are evaluated in Fig. 6. **a**, A held-out line's untreated cells and a desired treated population define the query; candidates comprise all 92 compounds. **b**, Mean reciprocal rank for the three scoring rules in Fig. 1a, with 95% bootstrap intervals: 0.945, 0.965 and 0.976. The two increments are +0.0201 and +0.0102. **c**, Top-1 decisions changed relative to each mean reference; ratios show the relative frequency of the two directions. **d**, The pooled +0.0303 advantage stratified by direction-only mean-cosine headroom; on 3,110 of 3,992 queries it already ranks the true drug first at every seed. **e**, Spearman correlation between direction-only mean-cosine headroom and population gain, with a within-context permutation null. **f**, Single-cell response detectability for a query drug versus vehicle cells from the same line, using up to 300 cells per arm; median accuracy 0.800 with labels and 0.555 without labels. **g**, Interaction statistics versus response magnitude and cells per arm. The naive $1 - \cos(r_1, r_2)$ gives -0.69 and -0.43 ; cross-fitted S_{int} gives $+0.66$ and $+0.10$. **h**, Standardised coefficients from regressions of energy-population reciprocal rank, with direction-only mean-cosine reciprocal rank as a covariate; the first three rows use $n = 3,618$ queries in 44 cell-line clusters and the naive cosine uses $n = 3,620$. Filled markers, intervals excluding zero.

cosine, almost the whole of energy-population retrieval's deficit, although it reads no distribution at all; against the magnitude-aware mean itself energy-population retrieval sits only -0.0020 below (-0.0036 to -0.0005 ; Fig. 6d). An additive predictor translates the query line's own vehicle cells by an effect averaged over the other 43 lines, so it reproduces that population's heterogeneity exactly while generating no drug-by-state interaction at all, an algebraic property of the model class rather than a training failure (Fig. 6e; Supplementary Note 4).

A state-conditioned average-effect predictor generates state-specific interaction while preserving

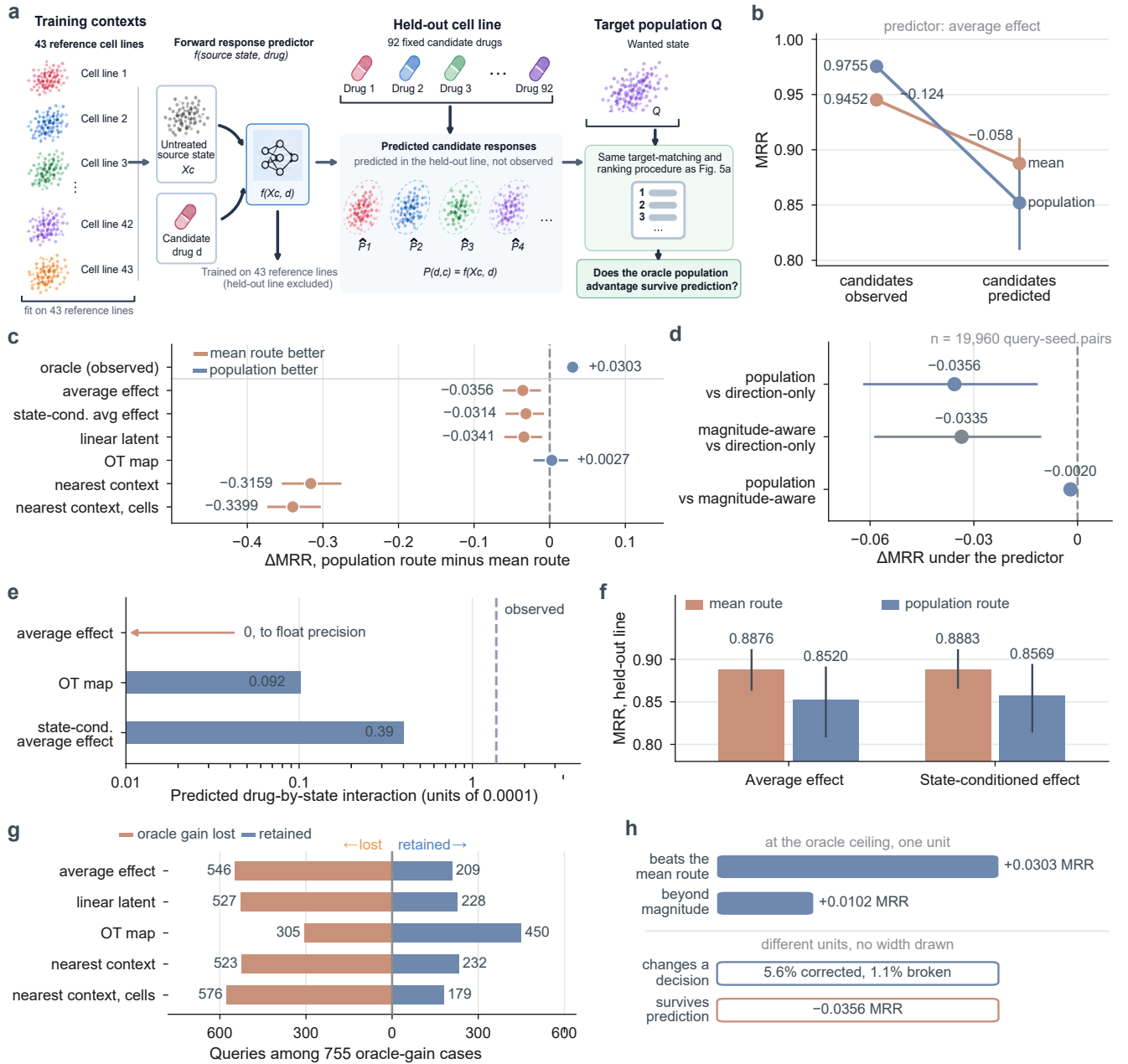


Figure 6: The observed-response population advantage is not preserved by the evaluated predictors. The task, queries, library and split are as in Fig. 5; candidate responses in the held-out line are produced by a forward predictor fitted on the other 43 lines. **a**, The intervention-identification task with predicted candidate responses. **b**, Both routes travelling from observed to predicted candidates under the primary average-effect predictor; direction-only mean cosine gives up 0.058 MRR and energy-population retrieval 0.124, and the two cross. **c**, ΔMRR of energy-population retrieval over direction-only mean cosine, at the oracle and for six predictors; under an additive predictor the oracle gain inverts rather than attenuating. **d**, The same predictions judged three ways: energy-population retrieval's -0.0356 deficit against direction-only mean cosine is matched by the magnitude-aware mean's -0.0335 , and -0.0020 remains once magnitude is controlled. **e**, Median drug-by-state interaction from each predictor versus the cross-fitted value measured in real cells (dashed); the additive predictor generates zero interaction. **f**, A state-conditioned average-effect predictor generates interaction while preserving direction-only mean-cosine accuracy, but energy-population retrieval remains lower. **g**, Among the 755 queries on which oracle energy-population retrieval had any gain, 209 retain a gain under the primary average-effect predictor and 546 do not. Additional gains occur among queries with no oracle gain: 343 for the primary predictor and 1,165 for the Gaussian OT map. **h**, MRR budget across Figs. 5 and 6; the first two rows share the oracle MRR scale, whereas the final two show a rate and a signed gain.

direction-only mean-cosine MRR (0.8883 against 0.8876 for the average-effect predictor), and it still leaves energy-population retrieval behind (-0.0314 , -0.0584 to -0.0076 ; Fig. 6f). State conditioning generated interaction, but did not restore the population-level advantage in this setting. The same attrition is visible query by query: of the 755 queries on which oracle energy-population retrieval had any gain, the average-effect predictor keeps one on 209 and loses it on 546 (Fig. 6g). Together the two arms locate where population-level information enters and where it is lost (Fig. 6h). With responses observed it is present, reaches the ranking decision, and is bounded by the performance of mean signatures. With responses predicted, the aggregate population-level advantage does not survive. An additive predictor preserved mean-level retrieval reasonably well while generating no state-specific differential response. State conditioning restored that interaction without restoring a population-level advantage.

Discussion

Our results separate two ideas that are often treated as equivalent: preserving more cellular detail and making a better intervention decision. Single-cell response populations retain variation that is necessarily lost by averaging, but only a fraction of this variation changes which drug is preferred. Across the settings examined here, much of the apparent advantage of population-level retrieval was already captured by the magnitude of the average response, leaving a smaller and less consistent contribution from cell-to-cell response structure. Related benchmarks of genetic perturbation prediction found that deep models did not consistently outperform simple linear baselines²⁰. Systematic variation can also inflate performance estimates in that setting²¹. In our study, the distinction became clearer as evaluation moved beyond transcriptomic response matching. Population-level gains did not consistently carry over to mechanism recovery or independent functional measurements, and even the same held-out protein measurements favoured different retrieval strategies depending on whether they were evaluated as average responses or as full single-cell populations. Thus, greater cellular resolution does not automatically translate into greater decision value. What matters is whether the additional information changes candidate ordering under an evaluation that reflects the intended biological use.

This distinction becomes more consequential when single-cell responses must be predicted rather than measured. In prospective use, a virtual-cell model must infer how each candidate drug would alter a cellular population that has not yet been experimentally perturbed. Zero-shot evaluations of scGPT and Geneformer have likewise found limitations in cell-type representations²². Our observed-response experiments show that population structure can contain information beyond the average response that is sufficient to change drug ranking. The prediction experiments, however, show that preserving cellular heterogeneity is not equivalent to preserving this decision-relevant information. Models can reproduce variation among cells without accurately predicting how a perturbation changes that variation, and recovering measurable state-dependent effects does not by itself guarantee that candidate ordering will improve. The relevant predictive target is therefore not the full response distribution in the abstract, but the components of that distribution that alter intervention choice. None of the predictors examined here simultaneously achieved strong generalisation to the target context and faithful recovery of the state-specific perturbation effects observed in real cells, leaving open whether more capable conditional models can preserve the population-level gain seen with observed responses.

More broadly, these results suggest that increasing biological resolution should be treated as a hypothesis about decision value rather than as an objective in itself. A higher-resolution representation is useful only if the additional information is reproducibly supported by the measurement, cannot be

recovered from a simpler representation, changes the downstream decision and can be preserved when prediction is required. These requirements separate four questions that are often conflated in virtual-cell evaluation: what biology is measurable, what information a representation retains, what information changes a decision and what information a model can predict in a new context. Their relative importance will depend on the perturbation library, cellular context, measurement modality and downstream objective, and the external criteria examined here capture biological or functional similarity rather than therapeutic benefit. Nevertheless, the broader principle is general: the goal of higher-resolution virtual-cell modelling should not be to reproduce every measurable aspect of cellular variation, but to identify and predict the variation that is both biologically supported and consequential for the decision the model is intended to inform.

Methods

Study design and data resources

We compared observed-population retrieval with mean signatures, tested fixed rankings against biological and functional criteria, measured natural response heterogeneity, and evaluated intervention identification with observed or predicted candidate responses. The primary observed-population retrieval analyses used SciPlex3⁶, comprising 276,325 cells across three cancer cell lines (A549, K562 and MCF7), 188 compounds shared across the three contexts and a 2,000-gene expression space. SciPlex3 supports both within-context retrieval and controlled cross-context constructions because the same compound library is measured across all three cell lines. Frangieh melanoma Perturb-CITE-seq¹⁶ spans Control, IFN γ and Co-culture conditions with 239 CRISPR knockouts. Both analyses used the same RNA matrix with different inclusion criteria: the retrieval analysis uses the 218,023 cells left after dropping nine knockout targets with fewer than 50 cells, and the evaluator-shape experiment uses all 218,331 cells of the RNA-protein barcode intersection, of which 215,915 contribute to at least one condition mean once the 60-cell floor per knockout is applied. Frangieh provides an independent perturbation-retrieval setting and matched surface-protein measurements for the evaluator-shape experiment, in which the protein modality is withheld from every retrieval method and used for evaluation. This is a genetic-perturbation test of the retrieval formulation.

Additional resources serve roles outside the primary retrieval benchmark. ZhaoSims2021⁷ contains drug-treated patient-derived glioblastoma tissue and supplies natural-tissue diagnostics of state-dependent responses and their recoverability. Tahoe-100M¹⁹ provides larger-scale measurements of the same information requirements under intrinsic within-cell-line heterogeneity and is the substrate for the source-to-target intervention-identification benchmark of Figs. 5 and 6. We analyse plate 3 only, which comprises 4,158,278 cells across 50 cell lines and 92 compounds plus vehicle, using the released 2,304-gene log1p-transformed representation without refitting the preprocessing pipeline. Finally, GDSC2 dose-response profiles provide an external functional readout for drug-drug similarity and are joined to the SciPlex3 compound panel independently of the transcriptomic retrieval scores. A CD34+ haematopoietic-cell subset²³ contributes only to 12 exploratory transfer tasks in Supplementary Note 2, not to the primary retrieval comparisons. Dataset scale, inclusion criteria and analysis roles are summarised in Supplementary Table 2.

Preprocessing and construction of perturbation-response populations

Single-cell expression analyses were performed in log-transformed expression space using the processed gene sets defined for each dataset. SciPlex3 and Frangieh were analysed in 2,000-gene representations.

For SciPlex3, cells were first capped at 120 per compound, cell line and dose and at 3,000 controls per cell line, using seed 0. From those retained cells, 60,000 cells were sampled without replacement with the same seed; raw counts were normalised to a total of 10^4 per cell, log1p-transformed, and 2,000 highly variable genes were selected using the Seurat highly-variable-gene criterion²⁴ as implemented in Scanpy²⁵. The final SciPlex3 matrix was then rebuilt from raw counts on those genes using all-gene size factors so that normalisation is applied once. For Frangieh, no HVG subsample was used: all cells passing the immune-condition and perturbation filters were normalised to 10^4 counts per cell, log1p-transformed, and reduced to 2,000 genes by the same Seurat criterion. For ZhaoSims2021, counts were normalised to a total of 10^4 per cell, log1p-transformed and reduced to 2,000 highly variable genes. Tahoe-100M was used in the released preprocessed form (2,304 highly variable genes, log1p transformed), with no additional feature selection or representation fitting. Dataset-specific filtering and minimum-cell requirements for individual analyses are given in the corresponding Methods sections below.

For a perturbation d measured in cellular context c , let

$$X_{d,c} = \{x_1^{(d,c)}, \dots, x_{n_{d,c}}^{(d,c)}\}$$

denote its treated single-cell expression profiles and let

$$\mu_{0,c} = \frac{1}{n_{0,c}} \sum_{j=1}^{n_{0,c}} x_j^{(0,c)}$$

denote the mean of the matched control population. The conventional perturbation signature is the control-referenced mean response

$$\bar{d}_{d,c} = \frac{1}{n_{d,c}} \sum_{i=1}^{n_{d,c}} x_i^{(d,c)} - \mu_{0,c}.$$

Mean-signature retrieval operates only on $\bar{d}_{d,c}$.

Mean-signature retrieval uses the control-referenced mean response, whereas population-level retrieval scores receive the raw normalised-expression treated-cell matrices,

$$P_{d,c} = \left\{ x_i^{(d,c)} \right\}_{i=1}^{n_{d,c}},$$

while variation among cells remains available to the population score. Mean cosine and mean L_2 each subtract the corresponding control mean from the treated-cell mean; query and candidate controls may therefore differ. Energy distance, RBF-MMD, sliced Wasserstein and coverage use the raw normalised-expression populations and do not subtract matched controls in the primary population comparisons. Matched-control-referenced population scores were evaluated separately as a sensitivity analysis (Supplementary Note 1). State- and compartment-specific responses use matched controls from the same biological context. Analysis-specific cell filters, sampling caps and seeds are given below; protein pre-processing is described with the evaluator-shape experiment.

Population-to-population retrieval and signature baselines

For query Q and candidate populations $\{P_d\}_{d=1}^D$, retrieval ranks the library by similarity or distance. Population scores use the empirical treated-cell populations; signature scores use their control-referenced mean responses. Each comparison uses the same queries and candidate library.

Population-level retrieval. We evaluated energy distance²⁶ (the primary population score), maximum mean discrepancy²⁷, sliced Wasserstein distance²⁸, and the mean and worst-case limits of a subpopulation-coverage score. All rank candidates from smallest to largest distance. For a candidate population $P = \{x_1, \dots, x_m\}$ and a query population $Q = \{y_1, \dots, y_n\}$ the energy distance is estimated as

$$\hat{\mathcal{E}}(P, Q) = \frac{2}{mn} \sum_{i,j} \|x_i - y_j\| - \frac{1}{m(m-1)} \sum_{i \neq i'} \|x_i - x_{i'}\| - \frac{1}{n(n-1)} \sum_{j \neq j'} \|y_j - y_{j'}\|,$$

using the Euclidean norm. Energy and RBF-MMD use unbiased U-statistics, including within the coverage score. The secondary PCA-latent energy score in Fig. 2a uses the V-statistic, with within-population denominators m^2 and n^2 ; sliced Wasserstein has no U/V choice. Populations with fewer than two sampled cells, including empty coverage components, receive distance 10^6 . Sampling caps and secondary-score settings are in Supplementary Methods.

Subpopulation-coverage score. The query is partitioned into $K = 2$ components. Construction labels are used when available; otherwise query cells are partitioned by k -means and candidate cells assigned to query components as specified in Supplementary Methods. Writing d_k for the base distance in component k , the coverage distance and ranking score are

$$D_\beta = \frac{1}{\beta} \left(\log \sum_{k=1}^K e^{\beta d_k} - \log K \right), \quad S_\beta = -D_\beta.$$

The base distance is energy, MMD or sliced Wasserstein. As $\beta \rightarrow 0$, D_β averages component distances; as $\beta \rightarrow \infty$, it approaches the worst-matched component. Coverage-mean and coverage-worst use these limits; for $K = 1$, coverage equals the global distance.

Mean-signature baselines. Mean-signature retrieval ranks the control-referenced vectors $\bar{d}_{d,c}$ by cosine similarity or Euclidean distance. The implemented CMap-style cosine is numerically identical to mean cosine on these vectors. We also evaluated the rank-based L1000 weighted connectivity score (WTCS)⁵ as a distinct signature baseline. For the PCA comparison, one 30-component basis was fitted per query to a uniform sample of up to 4,000 cells pooled from the query and all 43 candidates. This transductive fit used expression values without candidate-identity, ground-truth or subpopulation labels; query cells were disjoint from the correct candidate's cells. Control cells were excluded from fitting, and control means were projected through the fitted basis. Retrieval then compared latent-space means or populations.

Connection between mean and population retrieval. Under a shared or common reference, define the control-referenced response population $R_{d,c} = P_{d,c} - \mu_{0,c}$, and write it as

$$R = \mu + \lambda \epsilon, \quad E[\epsilon] = 0,$$

where $\mu = \bar{d}_{d,c}$ is the control-referenced mean response, ϵ represents within-population variation and λ controls its magnitude. As $\lambda \rightarrow 0$, the response population collapses to a point mass at μ , and population comparison therefore loses all information beyond the mean. For two point masses, energy distance is $2\|\mu_Q - \mu_d\|_2$, giving the same ranking as mean L_2 under a common control reference. Query and candidate controls can differ in the primary comparisons, so that equivalence does not hold globally.

386 Cosine additionally removes response magnitude. These limits describe the information retained by
387 each representation; WTCS and the population distances remain distinct scoring functions²⁹.

388 HIR-Bench construction

389 HIR-Bench is a controlled two-subpopulation latent-welfare construction used to establish when within-
390 population structure can change a preferred candidate. Each synthetic query has a majority subpop-
391 ulation of weight α and a minority subpopulation; candidate response centres range from balanced to
392 majority- or minority-specialist, with a conflict parameter controlling the extent to which the two sub-
393 populations prefer different candidates. Per-subpopulation utility is negative expected squared distance
394 to that subpopulation’s latent target. Let d_M be the candidate preferred by the majority and d_C the can-
395 didate preferred by the minority, with

$$A = u(d_M, 0) - u(d_C, 0), \quad B = u(d_C, 1) - u(d_M, 1).$$

396 The mean-welfare preference changes at

$$\alpha^* = \frac{B}{A + B}.$$

397 When $A + B > 0$, mean welfare prefers d_M above the boundary and d_C below it. The worst-
398 subpopulation optimum over the full library instead maximises $\min_k u(d, k)$.

399 The full experiment crossed $\alpha \in \{0.50, 0.70, 0.85, 0.95\}$, seven conflict levels from 0 to 1, 15 or 60
400 cells per subpopulation, Gaussian noise standard deviation 0.3 or 0.8, candidate-library sizes 20 or 100,
401 and three information conditions, observed populations, predicted means and predicted structures, at
402 20 seeds (672 parameter configurations; 13,440 instances). The predicted-mean condition replaces each
403 candidate population by cells sampled around its own mean, whereas the predicted-structure condition
404 uses a PCA reconstruction plus lower-variance noise. The observed, predicted-mean and predicted-
405 structure conditions therefore differ only in the within-population information made available to the
406 retriever. Full generator and transfer-analysis details are provided in Supplementary Note 2.

407 Response-matching and partial-observation evaluation

408 **Observed-population retrieval.** When complete candidate response populations were available, re-
409 trieval was evaluated over heterogeneous queries. Each query is a population built by combining two
410 response modes at mixing weight $\alpha \in \{0.5, 0.7, 0.9\}$, and the correct answer is the candidate that repro-
411 duces both modes at the same weight, built from the held-out half of the same cells so that no cell is
412 shared between a query and its answer. Two further candidates carry one mode each, and 40 distractor
413 populations complete a 43-candidate library.

414 The three tasks differ in what the two modes are. In the controlled SciPlex3 task they are the HDAC-
415 and JAK-associated response sources within a single cell line at 10 μ M, each pooling every compound
416 annotated to that class, so the two single-mode candidates are HDAC-only and JAK-only populations
417 and each distractor is an individual compound of another class. In the cross-line and Frangieh tasks
418 they are one perturbation observed in two biological contexts, two cell lines or Control and IFN γ , so the
419 two single-mode candidates are that perturbation observed in one context only and each distractor is a
420 different perturbation mixed across the same two contexts at the same weight; cross-line perturbations
421 pool all measured doses. The two single-mode candidates are each referenced to their own single-context
422 control, whereas the correct answer and every distractor are referenced to the α -mixed control. The cross-
423 line and Frangieh tasks use the most divergent perturbations passing a split-half reliability floor in both

contexts and a cross-context response cosine below 0.9: 12 drugs per cell-line pair and 6 knockouts. Their retrieval estimates therefore describe a panel selected for cross-context divergence.

Populations contain 400 cells in the controlled task and 200 in the other two, sampled without replacement where possible and with replacement otherwise. Queries within a task setting share one fixed cell split and vary by perturbation or mechanism, α and resampling seed. The controlled and cross-line tasks use ten seeds and Frangieh uses five; queries within a setting are repeated observations. The both-mode candidate’s rank is one plus the number of candidates with strictly higher scores. This rank defines Hit@1, Hit@5, MRR and nDCG@10. A single-mode candidate ranked first counts as a miss, including when it carries the correct perturbation in one context. Hit@1 therefore evaluates population composition as well as perturbation identity. The primary summary is an unweighted macro-mean over seven task–setting combinations. Query-weighted summaries and the tie check are given in Supplementary Methods.

Partial-observation protocol. Each score ranks the same partially observed candidate library once, and the resulting ranking is evaluated under response matching and mechanism recovery. Candidates are ordered by a label-blind stable sort; Hit@1 uses the best rank attained by a member of the query’s mechanism or target set. Three splits construct the partial observation, all on SciPlex3, and in each the hidden material never enters the candidate library³⁰. In *leave-drug-out*, one drug is hidden, the query is that drug’s own treated population (up to 200 cells) and the library is every other drug in the context. In *leave-MoA-out*, an entire annotated mechanism class is removed from the library and the query is one drug drawn from the removed class, so recovery requires a same-target surrogate among the remaining drugs. In *partial-library*, a fraction of the drugs is treated as measured, the query is built from the pooled responses of the unmeasured drugs, and the retriever selects the next batch of candidates. In all three, query states are obtained by k -means with $k = 2$ on the query cells and the minority state is the smaller cluster; candidate populations are capped at 120–150 cells per drug.

The response-matching criterion is an energy-based welfare regret. Each candidate’s welfare is the aggregate over query states s of its negative energy distance to that state, $U(d) = \text{agg}_s (- \mathcal{E}(P_d, Q_s))$ with the aggregator taken as the minimum in the worst-case form, and the regret of the selected candidate $\hat{d}$ is

$$\text{Regret}(\hat{d}) = \max_{d \in \mathcal{L}} U(d) - U(\hat{d}) \geq 0,$$

where $\mathcal{L}$ is the observed library. For population scorer p and mean reference r , regret reduction is $\text{Regret}(\hat{d}_r) - \text{Regret}(\hat{d}_p)$; positive values indicate improved response matching. Figures 2c and 3a compare worst-case coverage with mean cosine; Fig. 2e compares every population scorer with mean cosine and mean L_2 . The energy-based criterion evaluates response fidelity using information also available to the population scorer.

Task-proximal biological evaluation and validation of the state-similarity measure. Mechanism-of-action (MoA) recovery was measured by normalised discounted cumulative gain (nDCG)³¹, rewarding candidates with the query’s annotated mechanism near the top of the fixed ranking. Evaluation used the 600 leave-drug-out queries; the 120 leave-MoA-out and 45 partial-library queries were excluded because their libraries contain no member of the query mechanism class. Among the 600, 21 queries lacked a same-mechanism candidate and received nDCG = 0 for every method. They were retained as ties; the exclusion sensitivity is reported in Supplementary Methods.

We assessed a mean-based minority-state similarity score defined as $\frac{1}{2}\{1 + \cos(\bar{P}_d, \bar{Q}_{\min})\}$, where $\bar{P}_d$ is the candidate mean response and $\bar{Q}_{\min}$ is the mean response of the less represented query state. The cell-level reference was the fraction of minority-query cells whose Euclidean distance to their nearest candidate cell did not exceed the median distance pooled over all candidates in that seed. The comparison used a K562 HDAC–JAK controlled mixture ($\alpha = 0.7$), 20 fixed seeds and 43 candidates per seed (860 candidate cases). We compared the two metrics by pooled Spearman correlation and by agreement of the top-ranked candidate. The mean-based score did not agree with the cell-level measure and was not used as a biological endpoint.

External functional evaluation using GDSC2

We evaluated SciPlex3 rankings against drug–drug functional similarity derived from GDSC2 dose–response profiles^{17,18} (release 8.5; Data availability). SciPlex3 compounds were matched to GDSC2 by normalised drug name, with substring matching used only as a fallback. After removal of one duplicate mapping and restriction to the GDSC2 screening platform, 35 compounds remained. For each matched compound, its functional profile was the vector of fitted dose–response AUC values across GDSC2 cell lines. The three cell lines used for the transcriptomic retrieval experiments, A549, K562 and MCF7, were excluded from these profiles, leaving 966 cell lines.

Cell lines differ systematically in their overall sensitivity to drug treatment, which can induce positive correlations between otherwise unrelated compound profiles. We therefore removed this shared cell-line effect before calculating drug–drug similarity. For cell line l and compound d , the centred response was

$$\widetilde{\text{AUC}}_{d,l} = \text{AUC}_{d,l} - \frac{1}{|\mathcal{D}_l|} \sum_{d' \in \mathcal{D}_l} \text{AUC}_{d',l},$$

where $\mathcal{D}_l$ contains all GDSC2 compounds measured in cell line l , rather than only the 35 compounds overlapping SciPlex3. Functional similarity between query drug q and candidate drug c was then defined as

$$F(q, c) = \rho_s \left(\widetilde{\text{AUC}}_{q,\bullet}, \widetilde{\text{AUC}}_{c,\bullet} \right),$$

where ρ_s denotes Spearman correlation across cell lines shared by the two compounds. Drug pairs measured in fewer than 100 common cell lines were excluded. This yielded 591 usable drug pairs, with a median of 909 shared cell lines per pair. The corresponding uncentred functional similarity was retained as a sensitivity analysis.

For each SciPlex3 cell line, retrieval was evaluated in a leave-one-drug-out design. The 10 μM response population of the held-out compound served as the query, and the remaining matched compounds formed the candidate library, giving 30–34 evaluable candidates per query after valid-pair filtering. Each RNA-based retrieval rule produced a candidate ranking without access to GDSC2 data. Agreement between that ranking and the external functional ordering $F(q, c)$ was quantified by Spearman correlation across candidates. The primary RNA-derived rankings were energy-distance population retrieval and control-referenced mean-cosine retrieval. To test whether apparent functional agreement required cell-to-cell distributional information, we additionally used a response-magnitude matching score,

$$M(q, c) = - \left| \|\bar{d}_q\| - \|\bar{d}_c\| \right|,$$

which prefers candidates whose overall transcriptional response magnitude is similar to that of the query. A query-independent magnitude score $\|\bar{d}_c\|$ was retained as a diagnostic control. An uncorrected

mean-cosine score based on treated means without matched-control subtraction and a potency-matching diagnostic were included as sensitivity controls.

To estimate the component of the energy-based association not explained by response magnitude, we computed a partial rank correlation with the magnitude-matching channel removed. Retrieval scores and functional similarities were rank-transformed, each was residualised with respect to $M(q, c)$, and the correlation between the two residuals was reported as the magnitude-adjusted association. Overall values are median per-query Spearman correlations across 103 queries, with cell-line-specific medians as higher-level summaries. Queries share candidate pools. The pre-specified comparison of energy retrieval with the response-magnitude scalar reports pooled direction counts descriptively.

Protein-based evaluator-shape experiment

Frangieh Perturb-CITE-seq¹⁶ provides matched RNA and surface-protein measurements from the same cells, allowing the statistical form of an external evaluator to be changed while holding its biological measurement fixed. RNA and protein matrices were joined by cell barcode before analysis. Of the 24 measured antibodies, four isotype controls were removed after normalisation, leaving 20 surface-protein features. Antibody-derived tag counts³² were centred-log-ratio normalised within each cell, $y_{ij} = \ln(1 + x_{ij}) - m^{-1} \sum_k \ln(1 + x_{ik})$ for cell i and antibody j , with a pseudocount of one, a natural logarithm and the per-cell mean taken over all $m = 24$ measured antibodies including zero counts and the four isotype controls; normalisation therefore precedes isotype removal. All analyses were performed separately within the Control, IFN γ and Co-culture conditions to avoid mixing immune context with perturbation identity. Knockouts represented by at least 60 barcode-matched cells were retained, and populations were subsampled to at most 250 cells using a fixed random seed.

Protein-based evaluators. For knockout k in immune condition c , let $\bar{d}_{k,c}^{\text{prot}}$ denote its mean control-referenced protein response and let $D_{k,c}^{\text{prot}}$ denote the corresponding set of per-cell control-referenced protein responses. To separate evaluator resolution from score geometry, we defined three protein-side evaluators paralleling the RNA-side retrieval representations. The direction-only evaluator was

$$E_{\text{dir}}(q, d) = \cos \left(\bar{d}_{q,c}^{\text{prot}}, \bar{d}_{d,c}^{\text{prot}} \right).$$

The magnitude-aware mean evaluator was

$$E_{\text{mag}}(q, d) = - \left\| \bar{d}_{q,c}^{\text{prot}} - \bar{d}_{d,c}^{\text{prot}} \right\|_2.$$

The population evaluator was

$$E_{\text{pop}}(q, d) = -\mathcal{E} \left(D_{q,c}^{\text{prot}}, D_{d,c}^{\text{prot}} \right),$$

where $\mathcal{E}$ denotes the energy distance. Larger values therefore indicate a closer match for all three evaluators. The primary evaluator-resolution comparison contrasted E_{mag} and E_{pop} . These share the same Euclidean-distance geometry in the zero-variance limit, in which the energy distance between two point masses is proportional to the Euclidean distance between their means; E_{dir} was retained as a direction-only reference.

For all three definitions, protein responses were referenced to the matched control population from

534 the same immune condition. Thus,

$$\bar{d}_{k,c}^{\text{prot}} = \text{mean} \left(X_{k,c}^{\text{prot}} \right) - \mu_{0,c}^{\text{prot}},$$

535 and

$$D_{k,c}^{\text{prot}} = \left\{ x_{i,k,c}^{\text{prot}} - \mu_{0,c}^{\text{prot}} \right\}_{i=1}^{n_{k,c}}.$$

536 All evaluator forms use the same cells and matched control.

537 **Fixed RNA-derived rankings.** Direction-only mean cosine, magnitude-aware mean L_2 and energy-
538 distance RNA rankings were computed once and evaluated against each protein criterion. The primary
539 comparison uses mean L_2 and energy distance, with cosine as a directional reference. RNA popula-
540 tions within an immune condition share a matched control, whose subtraction leaves energy distances
541 unchanged. Queries, candidate libraries, sampled cells and RNA scores were fixed across protein evalu-
542 ators; protein measurements were used only for evaluation.

543 **Response-magnitude control.** The scalar response-magnitude control was

$$M(q, d) = - \left\| \left\| \bar{d}_{q,c}^{\text{RNA}} \right\| - \left\| \bar{d}_{d,c}^{\text{RNA}} \right\| \right\|.$$

544 The same magnitude-matching ranking was evaluated against all three protein criteria. Agreement was
545 measured by Spearman correlation across candidate knockouts within each query and summarised by
546 the median across all 659 queries, as well as separately within Control, IFN γ and Co-culture. Paired
547 energy-minus-mean- L_2 differences were calculated within each query before taking their median.

548 **Natural heterogeneity and large-scale validation**

549 Patient-derived glioblastoma and Tahoe-100M were used to measure state-specific responses, their re-
550 coverability and agreement of drug ordering across naturally occurring cellular states.

551 **Patient-derived glioblastoma.** The ZhaoSims2021 dataset⁷ contains acute-slice cultures and surgical
552 biopsies from ten glioblastoma patients exposed to six compounds: Ana-12, Ispenisib, RO4929097,
553 Tazemetostat, etoposide and panobinostat. Expression counts were normalised to 10^4 per cell, log1p-
554 transformed and reduced to 2,000 highly variable genes. Only one patient received more than two
555 compounds. We therefore measure state-specific response, recoverability and the extent to which
556 population-level similarity differs from compartment-specific response ordering.

557 **Cellular-compartment assignment.** Cells were assigned to malignant and myeloid compartments us-
558 ing canonical marker scores; ambiguous cells were excluded. Drug responses were defined relative to
559 patient- and compartment-matched controls. Assignment criteria, retained-cell counts and threshold-
560 sensitivity analyses are reported in Supplementary Note 3.

561 **Differential response in natural tissue.** State-specific perturbation responses were measured between
562 malignant and myeloid compartments. For patient p , drug d and compartment k , the response vector
563 was

$$r_{p,d,k} = \text{mean} \left(X_{p,d,k}^{\text{treated}} \right) - \text{mean} \left(X_{p,k}^{\text{control}} \right).$$

Only patient–drug combinations with at least 50 treated cells in both compartments were included. Controls were matched within both patient and compartment. Differential response was summarised as

$$R_{p,d} = \cos(r_{p,d,\text{malignant}}, r_{p,d,\text{myeloid}}).$$

This construction removes pre-existing differences between patients and cellular compartments before comparing the directions of their drug-induced responses.

Recoverability in natural tissue. Response-identity recoverability was assessed from within-patient drug-pair splits by comparing k -means, Gaussian-mixture, Leiden³³ and HDBSCAN³⁴ partitions with supervised classification. Patient-level bootstrap and method-pair sensitivity analyses are reported in Supplementary Note 3.

Proxy for downstream decision relevance. The glioblastoma cohort contains insufficient drug coverage for direct retrieval. We quantified whether whole-population drug similarity preserves the ordering observed within a specific cellular compartment. For each pair of drugs measured in the same patient, we first computed the cosine similarity between their population-mean response signatures and the cosine similarity between their malignant-compartment response vectors. Agreement between these two similarity structures was summarised by Spearman correlation across drug pairs.

That whole-population comparison gives a Spearman correlation of +0.878. Because the population-mean signature is the equal-weight average of the two compartment responses, the malignant compartment contributes half of it. We also used a disjoint comparison. For drug pair (a, b) , myeloid-response similarity was defined as

$$S_{a,b}^{\text{myeloid}} = \cos(r_{a,\text{myeloid}}, r_{b,\text{myeloid}}),$$

and malignant-response similarity as

$$S_{a,b}^{\text{malignant}} = \cos(r_{a,\text{malignant}}, r_{b,\text{malignant}}).$$

We then computed

$$\rho_{\text{disjoint}} = \rho_S(S_{a,b}^{\text{myeloid}}, S_{a,b}^{\text{malignant}}),$$

using only drug pairs available within the same patient. The two sides contain no shared cells and each compartment is referenced to its own matched control. This statistic measures the candidate- ordering headroom available to population structure.

Large-scale validation in Tahoe-100M. Tahoe-100M plate 3 was used to test the same requirements at larger scale. We evaluated two independently defined states, released cell-cycle labels and a control-derived k -means partition, and quantified state-specific response alignment, response-identity recoverability and agreement of drug ordering across disjoint states. Dataset filtering and all state-definition details are reported in Supplementary Note 3. For each eligible drug–cell-line condition, the response of state k was defined relative to its own state-matched controls,

$$r_{d,c,k} = \text{mean}(X_{d,c,k}^{\text{treated}}) - \text{mean}(X_{c,k}^{\text{control}}).$$

593 Differential response was measured as the cosine between the two state-specific response vectors,

$$R_{d,c} = \cos(r_{d,c,0}, r_{d,c,1}).$$

594 A condition was included only when at least 50 cells were available in each required treated and control
595 state.

596 Finally, a disjoint state-ordering statistic was used as a large-scale proxy for downstream decision
597 relevance. Within each cell line, cosine similarity was computed for every eligible pair of drug responses
598 separately in the two non-overlapping cellular states. Spearman correlation between the two resulting
599 drug-drug similarity matrices was then calculated,

$$\rho_c = \rho_s(S_{a,b}^{(0)}, S_{a,b}^{(1)}).$$

600 High correlation indicates that the two cellular states preserve similar relative ordering among drug
601 responses; lower correlation leaves greater scope for state-specific information to change candidate pref-
602 erence. This statistic is a proxy for decision relevance. Detailed recoverability methods are in Supple-
603 mentary Note 3.

604 **Source-to-target intervention identification on Tahoe-100M**

605 The Tahoe-100M plate 3 task identifies the compound that generated a desired treated population from
606 a starting vehicle population. Candidate responses are either observed in the target context or predicted
607 from other contexts. The outcome is the generating compound’s rank.

608 **Task definition and eligibility.** A context is one cell line. For a held-out context c and a query drug q ,
609 the retriever receives the vehicle (DMSO) cells of c and an observed treated population $X_{q,c}$, and ranks all
610 92 candidate compounds; the query drug’s identity q is hidden from the retriever. A condition entered
611 the grid if it had at least 100 treated and 200 vehicle cells, which admitted 3,992 eligible queries over 44
612 usable cell lines from the 4,600 conditions present on the plate. Each query was evaluated at five fixed
613 seeds (13, 29, 47, 71, 101), giving 19,960 query-seed pairs. Source and target populations were drawn
614 without replacement at up to 200 cells each, all available cells being taken where a condition held fewer,
615 and responses were referred to the source mean, $P = \{x_i - \bar{x}_{\text{source}}\}$.

616 **Oracle arm.** In the oracle arm every candidate’s response in the held-out context is taken from the
617 observed data, isolating the retrieval step and defining an information ceiling for prediction. Cell dis-
618 jointness is enforced through the library preparation: the target population is drawn from one sublibrary
619 half and every candidate bank from the other, at up to 100 cells per candidate, all cells in that half being
620 taken where a candidate has fewer, without replacement, so the generating drug is drawn from the same
621 half as its 91 competitors and is not penalized by a technical shift.

622 **Prediction arm.** Aside from the observed treated population that defines the target query, no treated
623 cells from the held-out context are used to fit the forward predictor or construct candidate-response pop-
624 ulations. Candidate responses are produced by a forward predictor fitted on the other 43 cell lines. Six
625 predictors were evaluated. Three are additive: they emit one signature per drug and apply it identically
626 to every source cell, so the candidate population is the vehicle population rigidly translated. These are
627 an average effect δ_d pooled over training contexts; a nearest-context transfer taking δ_d from the training

context whose vehicle mean is closest in cosine; and a linear-latent variant truncating the average effect to the principal components carrying 80% of training variance. Three allow cell-dependent responses. The *state-conditioned* predictor estimates the same average effect separately for each of the three cell-cycle states released with the plate, G1, S and G2M, and applies it to each source cell according to its state. Each per-state effect is averaged over the training contexts in which that state holds at least 20 cells; groups below the floor are omitted, and 359 of the 12,275 cell-line, drug and state groups fall below it, 294 of them in S phase. A drug left with no usable estimate for a state would keep its pooled effect there and be additive for that state alone; this did not occur for any of the $44 \times 92 \times 3$ state-specific response estimates. The predictor is non-additive over its whole output. The *Gaussian optimal-transport* map is a closed-form Bures map in a 50-component training principal-component basis. For source and treated latent covariances S_0 and S_1 , its linear map is

$$A = S_0^{-1/2} \left(S_0^{1/2} S_1 S_0^{1/2} \right)^{1/2} S_0^{-1/2}.$$

For a held-out source cell x , with P the loading matrix and $z = P^\top(x - \bar{x}_{\text{source}})$, the prediction is $\hat{x} = x + P\{(A - I)z + b_d\}$, where b_d is the average training-context treated-minus-vehicle latent mean shift for drug d . Vehicle and treated clouds are centred on their own condition means before pooling, so S_0 and S_1 estimate within-condition scatter. Each covariance is regularised by adding 10^{-3} times its mean diagonal element to the diagonal. Symmetric matrix square roots use eigendecomposition with eigenvalues floored at 10^{-10} . Each held-out-context basis is fitted on up to 50,000 training cells sampled without replacement using a generator keyed by purpose, seed 1 and held-out-context index; each context-drug fit uses up to 200 cells drawn with the corresponding fixed generator. This Gaussian/Bures predictor is distinct from the entropic Sinkhorn optimal-transport diagnostic³⁵ in Supplementary Note 4. Additional SciPlex3 predictor comparisons, including scGen fitted using scvi-tools³⁶ and entropic optimal transport computed with POT³⁷, are reported in Supplementary Note 4 and Supplementary Table 4. *Nearest context, cells* transfers the neighbouring context’s actual treated cells for the drug, shifted by the difference of the two vehicle means. This transfers the neighbouring context’s observed response distribution.

Scoring and evaluation. Candidates were scored by the three rules of Fig. 1a: a direction-only mean cosine, a magnitude-aware mean L_2 , and the population route, primarily the unbiased U-statistic energy distance, with MMD and sliced Wasserstein as secondary variants. The secondary scores use settings specific to this task: sliced Wasserstein over 200 random projections at 100 quantiles, and a single RBF bandwidth fitted once by the median heuristic on the candidate library and held fixed across queries. Mean reciprocal rank of the true intervention is the primary outcome; Hit@1, Hit@5 and Hit@10 are secondary. Intervals are 2,000-fold bootstrap intervals resampled over cell-line clusters, which are the independent units of this design. A decision is counted as corrected when the population route places the true drug first and the reference does not, and as broken in the reverse case.

Headroom, response detectability and the conditional bottleneck analysis

Headroom and its null. Headroom is defined per query as $H = 1 - \text{RR}_{\text{ref}}$, where the reference is direction-only mean-cosine retrieval, and the gain is $G = \text{RR}_{\text{pop}} - \text{RR}_{\text{ref}}$ on the same query. A Spearman correlation between H and G includes the shared reference rank and the reciprocal-rank ceiling. Within each cell line, the population route’s reciprocal ranks are randomly permuted across queries, preserving both marginal distributions, the $\text{RR} \leq 1$ ceiling and the cell-line clustering; G is then recomputed against

the unpermuted reference and the correlation retaken. Two hundred permutations at a fixed seed give the interval reported in Fig. 5e.

Response detectability. For each held-out cell line c and query drug q , equal numbers of cells treated with q and vehicle cells of c form a balanced two-class problem, at 300 per arm or at the size of the smaller pool where that is below 300, sampled at a fixed seed. Chance accuracy is 0.5. The supervised ceiling A_{sup} is the out-of-fold accuracy of a five-fold cross-validated logistic probe ($C = 1$) on those cells. The unsupervised value A_{unsup} is k -means with $k = 2$ on the same cells, scored by best-permutation accuracy. The labels are external, defined by the source well rather than by an expression-derived cell state. Supervised accuracy provides an empirical reference for response detectability.

Conditional bottleneck regression. The model in Fig. 5h uses population-retrieval reciprocal rank as the outcome and reference mean-cosine reciprocal rank as a covariate. The explanatory variables are interaction ratio, interaction reproducibility $R_{\text{int}} = \cos(I_A, I_B)$ and response detectability A_{sup} ; a separate model uses the naive response cosine $1 - \cos(r_1, r_2)$. Conditioning on reference rank accounts for the shared reference in the gain statistic. Estimation used ordinary least squares with cell-line-clustered standard errors. Predictors were standardised to zero mean and unit variance, and complete-case analysis retained 3,618 queries.

Cross-fitted drug-by-state interaction

For the released G1 and G2M states ($s = 1, 2$), let r_s be the treated-minus-vehicle mean response in state s . State-dependent response is measured by $I = r_1 - r_2$. Cross-fitting separates reproducible interaction from sampling noise, which inflates both $\|I\|$ and $1 - \cos(r_1, r_2)$ for weak responses. Each of the four arms is therefore split into disjoint halves A and B , the interaction is estimated once in each, and the two estimates are multiplied:

$$S_{\text{int}} = \frac{\langle I_A, I_B \rangle}{p}, \quad S_{\text{main}} = \frac{\langle M_A, M_B \rangle}{p}, \quad M = \frac{1}{2}(r_1 + r_2),$$

with p the number of genes. Independent halves give $E[S_{\text{int}}] = 0$ under the additive null $E[I] = 0$. Both treated and vehicle arms are split to keep control noise independent between halves. Splits use disjoint library preparations when at least two sublibraries are available, and random cell halves otherwise. Arms were capped at 400 cells and required at least 50, and ten A/B splits were averaged per query.

Two derived quantities are used. The *interaction ratio*, $S_{\text{int}}/(S_{\text{int}} + S_{\text{main}})$, is computed only when $S_{\text{int}} + S_{\text{main}} > 0$ and is otherwise recorded as missing; no clipping or stabilization is applied. It is therefore not a bounded fraction: when defined it can be negative or exceed one. Two of the 3,620 scored Tahoe conditions have a non-positive denominator and are excluded by this rule, giving the 3,618 complete cases in Fig. 5h. The ratio uses cross-fitted numerator and denominator. $R_{\text{int}} = \cos(I_A, I_B)$ asks whether the interaction reproduces across halves. As a positive control, synthetic conditions were built from the plate's own cells by taking the G1 cells from one drug and the G2M cells from another, so interaction is present by construction in material with the same states and arm sizes as the queries. The naive cosine $1 - \cos(r_1, r_2)$ is a descriptive comparator in Fig. 5g and the sensitivity regression. Response cosines in the predictor and tissue analyses likewise describe directional alignment; the cross-fitted statistics above estimate reproducible interaction.

704 Statistical analysis

705 Statistical units were defined by the experimental design: cells estimate a perturbation-response popu-
706 lation, and retrieval performance is summarised at the designated cluster or task level³⁸. The partial-
707 observation arm uses the query drug as the primary inferential cluster. Of the 765 queries, 720 hold
708 out one of 143 drugs and the remaining 45 pool the unmeasured drugs of the partial-library split into a
709 single batch query; those pooled queries remain in descriptive summaries but are excluded from formal
710 drug-cluster inference. The 600 leave-drug-out queries on which mechanism recovery is scored come
711 from 130 drugs. Each drug is scored across three shared cell lines, several library fractions and up to five
712 random seeds, giving 417 distinct drug-by-cell-line contexts in the individual-drug analysis. Primary
713 95% intervals for this arm use 10,000 percentile bootstrap resamples of query-drug clusters, retaining all
714 rows belonging to each sampled drug. For paired comparisons, repeated observations are first collapsed
715 to one median effect per drug and a two-sided drug-level Wilcoxon signed-rank test is used when re-
716 ported. For the selected-versus-unselected comparison, inference is summarised by the cluster-bootstrap
717 interval because drugs can contribute to both groups.

718 Paired comparisons between retrieval methods outside the clustered partial-observation analysis
719 used two-sided Wilcoxon signed-rank tests; independent groups used Mann–Whitney U tests; mono-
720 tonic associations used Spearman rank correlation. Across pre-specified divergence strata, drug-level P
721 values were adjusted by the Benjamini–Hochberg procedure³⁹. Effect sizes accompany all tests. The
722 highest-divergence mechanism-recovery estimate and its uncertainty are reported in Supplementary
723 Note 1.

724 For the GDSC2 functional analysis, Spearman correlations were computed per query across its can-
725 didate drugs. Overall values pool 103 queries descriptively; within-cell-line summaries retain the three
726 cell lines as higher-level units, and query-level paired tests are nominal. The protein evaluator-shape
727 experiment was summarised within the Control, IFN γ and Co-culture conditions, its three higher-level
728 units. For HIR-Bench, cross-validation was grouped by the label-determining (α , conflict) pair, giving 28
729 effective parameter cells, and performance was summarised by pooled out-of-fold AUC. For the patient-
730 derived glioblastoma analysis, uncertainty in the recoverability gap was estimated by cluster bootstrap
731 at the patient level with 5,000 resamples, differences being taken within each drug-pair split before ag-
732 gregation; four patients contributed evaluable splits. Intervention-retrieval intervals are 2,000-fold boot-
733 strap resamples over cell-line clusters. Tahoe-100M summaries are taken across eligible drug–cell-line
734 conditions or within-cell-line drug pairs according to the statistic, and both cellular-state definitions,
735 released cell-cycle state and the control-derived k -means partition, were analysed in parallel.

736 Analyses were implemented in Python. Single-cell data handling used Scanpy²⁵; the k -means,
737 Gaussian-mixture, principal-component and logistic-regression procedures named above are the
738 scikit-learn implementations⁴⁰.

739 Data availability

740 The original source datasets are publicly available as described below. SciPlex3, Frangieh Perturb-
741 CITE-seq and ZhaoSims2021 were obtained in harmonised form from scPerturb⁴¹, Zenodo record
742 13350497 ([doi:10.5281/zenodo.13350497](https://doi.org/10.5281/zenodo.13350497), release 1.4). The exact harmonised files used in the anal-
743 yses were SrivatsanTrapnell2020_sciplex3.h5ad for SciPlex3, FrangiehIzar2021_RNA.h5ad and
744 FrangiehIzar2021_protein.h5ad for Frangieh Perturb-CITE-seq, and ZhaoSims2021.h5ad for the
745 patient-derived glioblastoma analysis. The primary SciPlex3 data are available from the Gene Expression
746 Omnibus under accession GSE139944⁶; the original Frangieh and ZhaoSims2021 datasets are described

747 in the corresponding publications^{7,16}. The CD34+ subset used in the exploratory transfer analysis is from
748 GEO accession GSE306429²³.

749 Tahoe-100M¹⁹ was released by Vevo Therapeutics and the Arc Institute under CC0 and is distributed
750 through the Arc Virtual Cell Atlas and Hugging Face (tahoebio/Tahoe-100M). This study uses plate 3
751 only, together with the released obs_metadata and sample_metadata tables. Analyses use the released
752 preprocessed expression matrix comprising 4,158,278 cells and 2,304 highly variable genes in log1p-
753 transformed space.

754 The external functional evaluation used GDSC2 fitted dose-response data from the Genomics of
755 Drug Sensitivity in Cancer, release 8.5 (GDSC2_fitted_dose_response_27Oct23.xlsx), available from
756 <https://www.cancerrxgene.org>. The compound-matching table used to align the SciPlex3 and GDSC2
757 drug panels is described in Methods.

758 The three core processed matrices and associated annotation tables are publicly available at <https://huggingface.co/datasets/Boom5426/PopRetrieve>.
759 Selected result tables are available in the code repository. The release contents and external input requirements are documented in the repository
760 README and data card.

762 Code availability

763 The PopRetrieve software and released experiment drivers are publicly available at <https://github.com/Boom5426/PopRetrieve>.
764 The repository documents the analyses and outputs included in the release.

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770 Competing interests

771 The authors declare no competing interests.

772 Additional information

773 Supplementary Information is available for this paper: Supplementary Methods, Supplementary Notes
774 1–4 and Supplementary Tables 1–5.

775 References

- 776 [1] Charlotte Bunne, Yusuf Roohani, Yanay Rosen, et al. How to build the virtual cell with artificial
777 intelligence: priorities and opportunities. *Cell*, 187(25):7045–7063, 2024. doi: 10.1016/j.cell.2024.11.
778 015.
- 779 [2] Haotian Cui, Chloe Wang, Hassaan Maan, Kuan Pang, Fengning Luo, Nan Duan, and Bo Wang.
780 scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature*
781 *Methods*, 21(8):1470–1480, 2024. doi: 10.1038/s41592-024-02201-0.
- 782 [3] Minsheng Hao et al. Large-scale foundation model on single-cell transcriptomics. *Nature Methods*,
783 21(8):1481–1491, 2024. doi: 10.1038/s41592-024-02305-7.
- 784 [4] Justin Lamb et al. The Connectivity Map: using gene-expression signatures to connect small
785 molecules, genes, and disease. *Science*, 313(5795):1929–1935, 2006. doi: 10.1126/science.1132939.

- 786 [5] Aravind Subramanian et al. A next generation Connectivity Map: L1000 platform and the first
787 1,000,000 profiles. *Cell*, 171(6):1437–1452.e17, 2017. doi: 10.1016/j.cell.2017.10.049.
- 788 [6] Sanjay R. Srivatsan et al. Massively multiplex chemical transcriptomics at single-cell resolution.
789 *Science*, 367(6473):45–51, 2020. doi: 10.1126/science.aax6234.
- 790 [7] Wenting Zhao et al. Deconvolution of cell type-specific drug responses in human tumor tissue with
791 single-cell RNA-seq. *Genome Medicine*, 13(1):82, 2021. doi: 10.1186/s13073-021-00894-y.
- 792 [8] Sydney M. Shaffer et al. Rare cell variability and drug-induced reprogramming as a mode of cancer
793 drug resistance. *Nature*, 546(7658):431–435, 2017. doi: 10.1038/nature22794.
- 794 [9] Zhiting Wei et al. Benchmarking algorithms for generalizable single-cell perturbation response
795 prediction. *Nature Methods*, 23(2):451–464, 2026. doi: 10.1038/s41592-025-02980-0.
- 796 [10] Mohammad Lotfollahi, F. Alexander Wolf, and Fabian J. Theis. scGen predicts single-cell perturba-
797 tion responses. *Nature Methods*, 16(8):715–721, 2019. doi: 10.1038/s41592-019-0494-8.
- 798 [11] Charlotte Bunne et al. Learning single-cell perturbation responses using neural optimal transport.
799 *Nature Methods*, 20(11):1759–1768, 2023. doi: 10.1038/s41592-023-01969-x.
- 800 [12] Mohammad Lotfollahi et al. Predicting cellular responses to complex perturbations in high-
801 throughput screens. *Molecular Systems Biology*, 19(6):e11517, 2023. doi: 10.15252/msb.202211517.
- 802 [13] Leon Hetzel, Simon Boehm, Niki Kilbertus, Stephan Günemann, Mohammad Lotfollahi, and
803 Fabian J. Theis. Predicting cellular responses to novel drug perturbations at a single-cell resolu-
804 tion. In *Advances in Neural Information Processing Systems*, volume 35, pages 26711–26722, 2022. doi:
805 10.52202/068431-1937.
- 806 [14] Yusuf Roohani, Kexin Huang, and Jure Leskovec. Predicting transcriptional outcomes of novel
807 multigene perturbations with GEARS. *Nature Biotechnology*, 42(6):927–935, 2024. doi: 10.1038/
808 s41587-023-01905-6.
- 809 [15] Reem Alsulami et al. Predicting and interpreting cell-type-specific drug responses in the small-
810 data regime using inductive priors. *Nature Machine Intelligence*, 8(3):461–473, 2026. doi: 10.1038/
811 s42256-026-01202-2.
- 812 [16] Chris J. Frangieh et al. Multimodal pooled Perturb-CITE-seq screens in patient models define
813 mechanisms of cancer immune evasion. *Nature Genetics*, 53(3):332–341, 2021. doi: 10.1038/
814 s41588-021-00779-1.
- 815 [17] Wanjuan Yang et al. Genomics of Drug Sensitivity in Cancer (GDSC): a resource for therapeutic
816 biomarker discovery in cancer cells. *Nucleic Acids Research*, 41(D1):D955–D961, 2013. doi: 10.1093/
817 nar/gks1111.
- 818 [18] Francesco Iorio et al. A landscape of pharmacogenomic interactions in cancer. *Cell*, 166(3):740–754,
819 2016. doi: 10.1016/j.cell.2016.06.017.
- 820 [19] Jesse Zhang et al. Tahoe-100M: mapping drug-induced molecular phenotypes at single-cell resolu-
821 tion. *Cell*, 189(19):5945–5961.e8, 2026. doi: 10.1016/j.cell.2026.08.035.

- [20] Constantin Ahlmann-Eltze, Wolfgang Huber, and Simon Anders. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nature Methods*, 22(8):1657–1661, 2025. doi: 10.1038/s41592-025-02772-6.
- [21] Ramon Viñas Torné et al. Systema: a framework for evaluating genetic perturbation response prediction beyond systematic variation. *Nature Biotechnology*, 44(6):1050–1059, 2026. doi: 10.1038/s41587-025-02777-8.
- [22] Kasia Z. Kedzierska, Lorin Crawford, Ava P. Amini, and Alex X. Lu. Zero-shot evaluation reveals limitations of single-cell foundation models. *Genome Biology*, 26(1):101, 2025. doi: 10.1186/s13059-025-03574-x.
- [23] Benjamin DeMeo et al. Active learning framework leveraging transcriptomics identifies modulators of disease phenotypes. *Science*, 390(6776):ead8577, 2025. doi: 10.1126/science.adi8577.
- [24] Rahul Satija, Jeffrey A. Farrell, David Gennert, Alexander F. Schier, and Aviv Regev. Spatial reconstruction of single-cell gene expression data. *Nature Biotechnology*, 33(5):495–502, 2015. doi: 10.1038/nbt.3192.
- [25] F. Alexander Wolf, Philipp Angerer, and Fabian J. Theis. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biology*, 19(1):15, 2018. doi: 10.1186/s13059-017-1382-0.
- [26] Gábor J. Székely and Maria L. Rizzo. Energy statistics: a class of statistics based on distances. *Journal of Statistical Planning and Inference*, 143(8):1249–1272, 2013. doi: 10.1016/j.jspi.2013.03.018.
- [27] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13(25):723–773, 2012.
- [28] Julien Rabin, Gabriel Peyré, Julie Delon, and Marc Bernot. Wasserstein barycenter and its application to texture mixing. In *Scale Space and Variational Methods in Computer Vision (SSVM 2011)*, pages 435–446, 2012. doi: 10.1007/978-3-642-24785-9_37.
- [29] Yuge Ji, Tessa D. Green, Stefan Peidli, Mojtaba Bahrami, Meiqi Liu, Luke Zappia, Karin Hrovatin, Chris Sander, and Fabian J. Theis. Optimal distance metrics for single-cell RNA-seq populations. *bioRxiv*, 2023. doi: 10.1101/2023.12.26.572833.
- [30] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. doi: 10.1016/j.patter.2023.100804.
- [31] Kalervo Järvelin and Jaana Kekäläinen. Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems*, 20(4):422–446, 2002. doi: 10.1145/582415.582418.
- [32] Marlon Stoeckius, Christoph Hafemeister, William Stephenson, Brian Houck-Loomis, Pratip K. Chattopadhyay, Harold Swerdlow, Rahul Satija, and Peter Smibert. Simultaneous epitope and transcriptome measurement in single cells. *Nature Methods*, 14(9):865–868, 2017. doi: 10.1038/nmeth.4380.
- [33] V. A. Traag, L. Waltman, and N. J. van Eck. From Louvain to Leiden: guaranteeing well-connected communities. *Scientific Reports*, 9(1):5233, 2019. doi: 10.1038/s41598-019-41695-z.

- 858 [34] Ricardo J. G. B. Campello, Davoud Moulavi, and Joerg Sander. Density-based clustering based on
 859 hierarchical density estimates. In *Advances in Knowledge Discovery and Data Mining (PAKDD 2013)*,
 860 pages 160–172, 2013. doi: 10.1007/978-3-642-37456-2_14.
- 861 [35] Marco Cuturi. Sinkhorn distances: lightspeed computation of optimal transport. In *Advances in*
 862 *Neural Information Processing Systems*, volume 26, pages 2292–2300, 2013.
- 863 [36] Adam Gayoso et al. A Python library for probabilistic analysis of single-cell omics data. *Nature*
 864 *Biotechnology*, 40(2):163–166, 2022. doi: 10.1038/s41587-021-01206-w.
- 865 [37] Rémi Flamary et al. POT: Python Optimal Transport. *Journal of Machine Learning Research*, 22(78):
 866 1–8, 2021.
- 867 [38] Jordan W. Squair et al. Confronting false discoveries in single-cell differential expression. *Nature*
 868 *Communications*, 12(1):5692, 2021. doi: 10.1038/s41467-021-25960-2.
- 869 [39] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: a practical and powerful
 870 approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300, 1995. doi:
 871 10.1111/j.2517-6161.1995.tb02031.x.
- 872 [40] Fabian Pedregosa et al. Scikit-learn: machine learning in Python. *Journal of Machine Learning Re-*
 873 *search*, 12:2825–2830, 2011.
- 874 [41] Stefan Peidli et al. scPerturb: harmonized single-cell perturbation data. *Nature Methods*, 21(3):
 875 531–540, 2024. doi: 10.1038/s41592-023-02144-y.